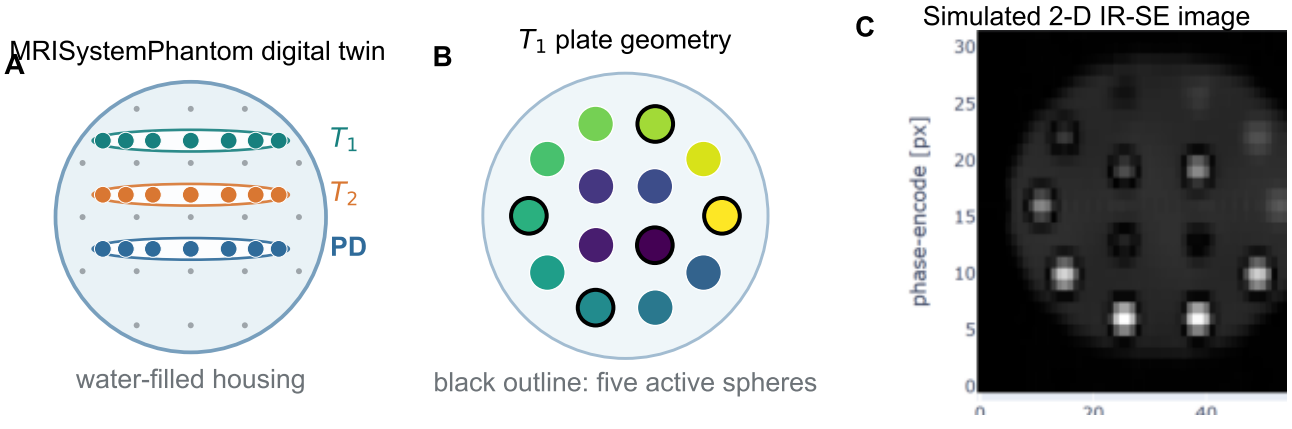
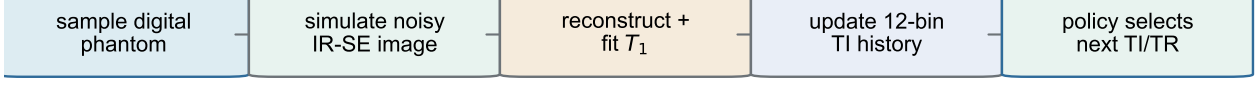

A
MRISystemPhantom digital twin
T_1
T_2
PD
water-filled housing
B
T_1 plate geometry
black outline: five active spheres
C
Simulated 2-D IR-SE image
phase-encode [px]
30
25
20
15
10
5
0
0
20
40

D


sample digital phantom
simulate noisy IR-SE image
reconstruct + fit T_1
update 12-bin TI history
policy selects next TI/TR